

Project initialization and planning phase

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 2 July 2026                                                  |
| Team Id       | XXXXX                                                        |
| Project Name  | Opti Crop: Smart Agricultural Production Optimization engine |
| Maximum marks | 3 marks                                                      |

Define problem Statement (customer Problem statements Template):

Smart Agricultural Production Optimization Engine addresses this challenge by using machine learning to analyze soil and environmental parameters and recommend the most suitable crop for cultivation. The system provides accurate, data-driven crop recommendations, helping farmers improve productivity, maximize profits, optimize resource utilization, and promote sustainable agricultural practices.

Example:

I am

A farmer who wants to improve crop production.

I'm trying to

Find the most suitable crop for my land using soil and climate data.

But

I don't have accurate information or expert knowledge.

Because

Soil nutrients, climate and environmental conditions vary and are complex to analyze.

Which makes me feel

Uncertain and worried about low yield and financial loss.

| Problem Statement (PS) | I am (Customer)            | I'm trying to                                   | BUT                                                                           | Because                                                                      | Which makes me feel                               |
|------------------------|----------------------------|-------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------|
| PS-1                   | A farmer                   | Select the most suitable crop for my land.      | I don't know which crop is best for my soil and weather conditions .          | I have limited knowledge of soil nutrients and changing climatic conditions. | Confused and worried about crop yield and income. |
| PS-2                   | An agricultural researcher | Analyze crop and environmental data accurately. | It is difficult to identify useful patterns from large agricultural datasets. | Manual analysis is time-consuming and complex.                               | Frustrated and less productive.                   |
| PS-3                   | An agricultural officer    | Recommend suitable crops to farmers.            | I don't have a reliable system to provide accurate crop recommendations.      | Crop selection depends on multiple environmental factors.                    | Concerned about giving incorrect recommendations. |